



AI Consulting Project for CBEC comparing Naïve Bayes and Non-Naive Networks to Mitigate Limitations in Collected Data

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Abstract:

Our research project focuses on using AI to evaluate the efforts of the Cell Biology Education Consortium (CBEC), an education initiative funded by the NSF to incorporate cell culture-based research into the classroom. The project explores how machine learning techniques can be leveraged to evaluate an education initiative funded by a grant. We explore the potential of Bayesian networks as a tool for modeling and evaluating CBEC user behavior using Wix website analytics data to train and test our models. We compared two different Bayesian network models, a naive Bayes approach and a causal network. As a proxy for CBEC user interest, the target variable we used for these networks is the duration of a given user session on the website, with the hopes of identifying useful indicators of increased website engagement. Each model considers the same variables but structures the respective networks differently. K-Fold validation was used to evaluate the performance of our Bayesian models developed using data collected from CBEC's website.

Background:

Bayesian Network:

A Bayesian network is a probabilistic graphical model that represents a set of variables and their conditional dependencies using directed acyclic graphs. Bayesian networks are widely used in fields such as machine learning, artificial intelligence, and decision analysis. They make use of some different probability formulas:

$P(A, B) = P(A B)P(B)$	Chain rule
$P(B A) = \frac{P(A B)}{P(A)} P(B)$	Bayes' Theorem
$P(A) = \sum P(A B_n)P(B_n)$	Law of total probability
$P(X_n X_1, \dots, X_{n-1}) = P(X_n Parents(X_n))$	Total probability in a Bayesian network

Computations:

We performed calculations on our laptops (MacBook Air 8GB memory, 8 core CPU) using python as the programming language to construct Bayesian networks. The following software packages were utilized to construct the networks and make queries to the model.

- Scikit-learn package
- Pgmpy package

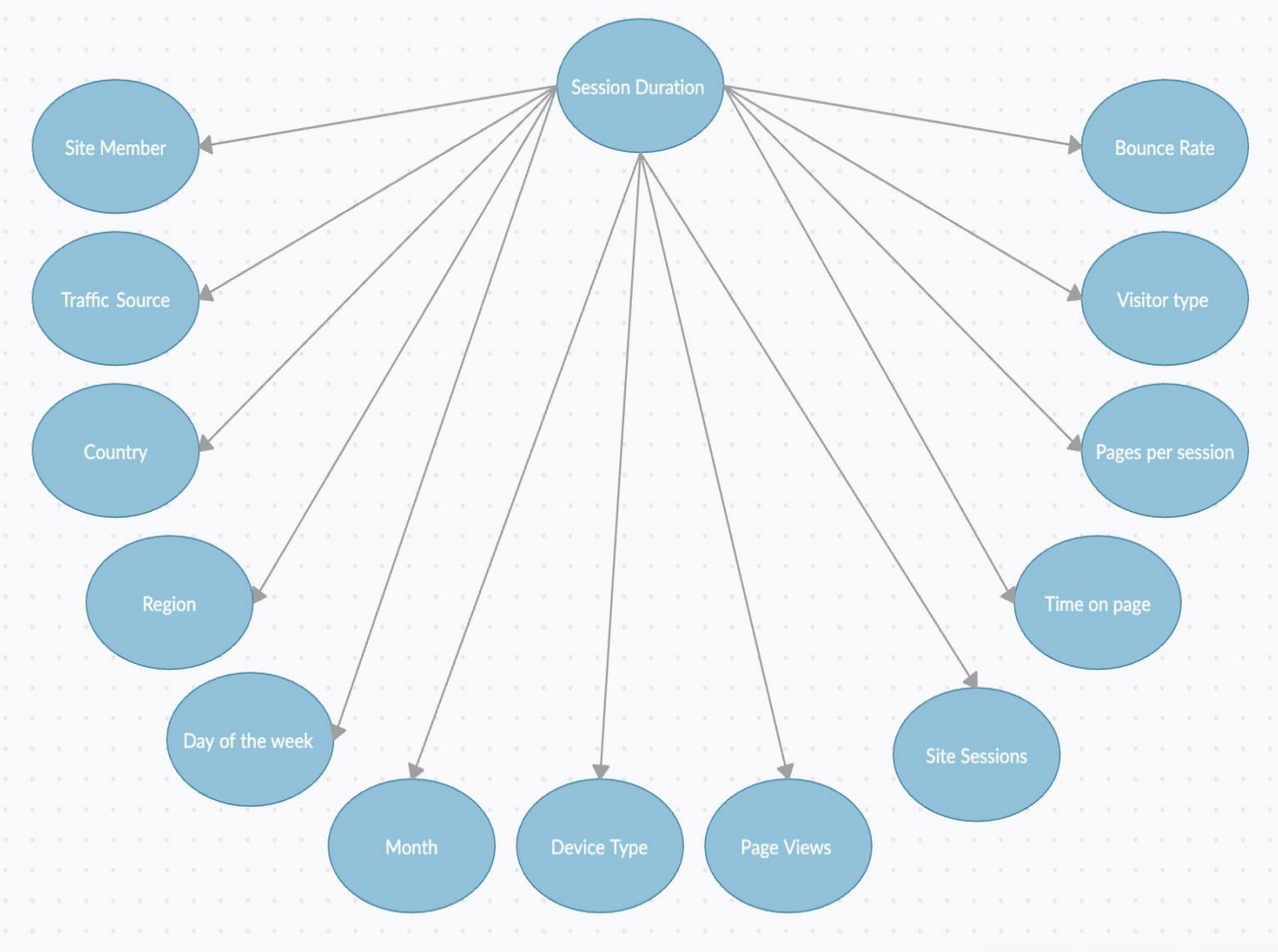
Dataset:

Data was collected from Wix analytics, but required extensive data cleaning and preprocessing before analysis could be conducted. From January 2018 to April 2023, there were a total of 3369 values in the dataset. We worked with the networks consisting of 13 variables, including session duration, average time spent on the website's page, country, region, and others.

Methodology:

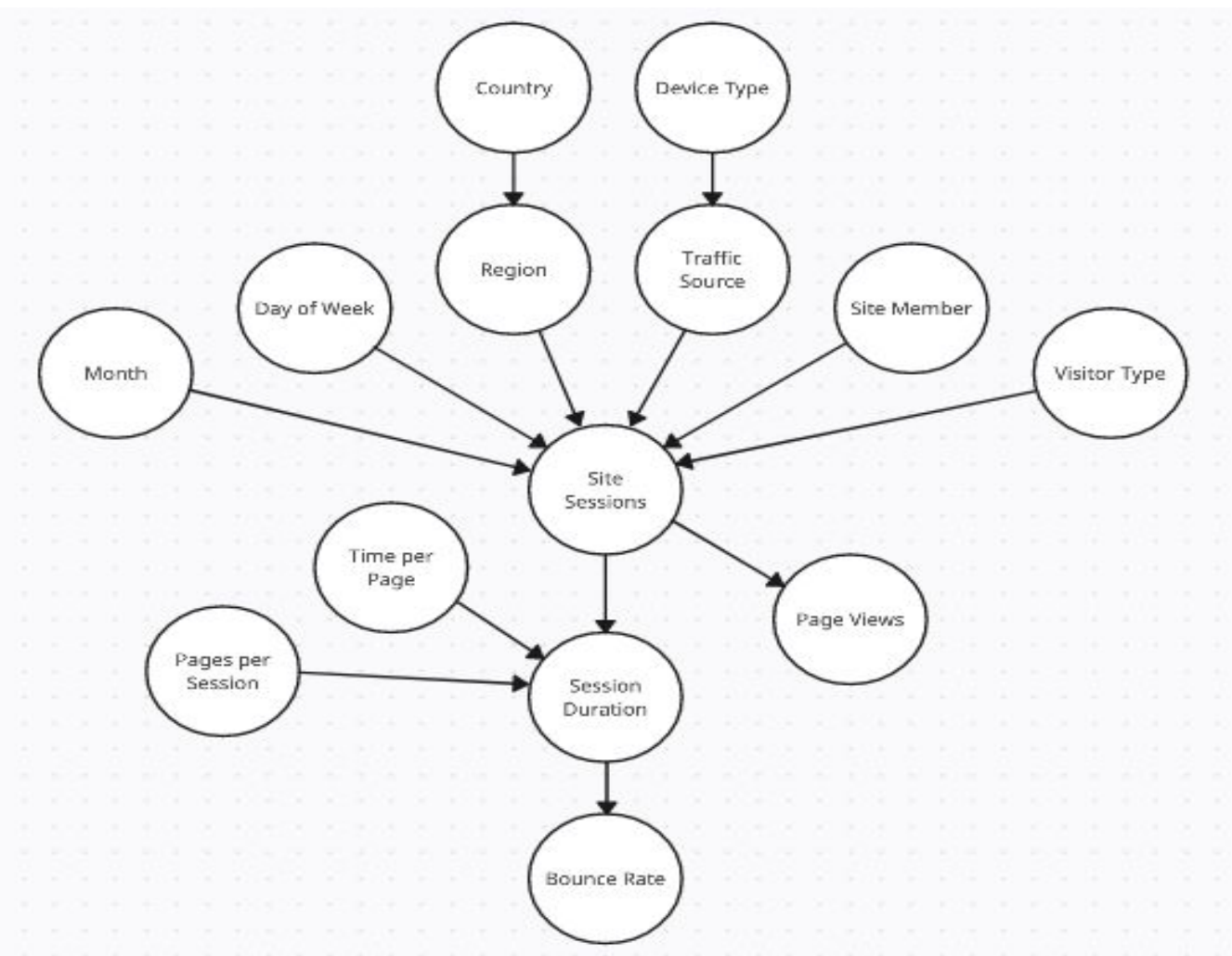
- Used Python to create Bayesian Network(s).

Naïve Bayes Model (Fig. 1) Probabilistic classification model based on Bayes' theorem and the assumption of independence between features. Naive Bayes can achieve high accuracy and requires relatively little training data.



(Fig. 1)

Causal network (Fig. 2): Graphical model that represents causal relationships between variables, where one variable influences the state of another. It allows for modeling the impact of interventions or changes in one variable on other variables in the network.



(Fig. 2)

- Imported data from using Wix.com and refined the webpage data collected.
- Used the variables from the data in the Bayesian network to fit the data.
- Utilized the Bayesian network model to investigate questions such as:

P(Session Duration | Region = South)

Results from the Naïve Bayes network:

Session_Duration	phi(Session_Duration)
Session_Duration(0-2)	0.8611
Session_Duration(3-7)	0.0669
Session_Duration(>7)	0.0720

Results from Causal network:

Session_Duration	phi(Session_Duration)
Session_Duration(0-2)	0.8987
Session_Duration(3-7)	0.0494
Session_Duration(>7)	0.0519

- Used K-Fold validation to evaluate the accuracy of each model.

K-Fold Validation:

We were able to assess the performance of the models on a range of different subsets of the data, which helped to identify any potential overfitting or underfitting issues.

For Naïve Bayes Network

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Accuracies: [0.87834, 0.88724, 0.91395, 0.86053, 0.89021, 0.87537, 0.89614, 0.92285, 0.89583, 0.86607]
Mean: 0.88865
SD: 0.01872
```

For Causal Network

```
Accuracies: [0.87537, 0.87834, 0.89911, 0.87537, 0.90208, 0.89318, 0.88131, 0.89318, 0.89881, 0.89286]
Mean: 0.88896
SD: 0.00982
```

Conclusion:

Using the K-Fold validation method with 10 folds showed that both models achieved nearly 90% accuracy for predicting the duration of a given user session. Neither model seemed to clearly outperform the other, which may be a result of the limited data we were able to work with. Using these models, we can also answer questions like:

On what weekday(s) are users more likely to have longer sessions on the website?

According to the causal model, Saturday & Sunday had the greatest probabilities of session durations longer than 7 minutes.

Future Work:

We would like to see how well the causal network model compares versus a neural network at predicting the probability of a given state of the target variable. Additionally, as CBEC grows and sees more engagement, we would like to see how these models perform with more data points. One future project may also include the development of a new CBEC website with code implemented to gather better data than was available via Wix analytics.

References:

- Russell, Stuart J. (Stuart Jonathan), 1962-. *Artificial Intelligence : a Modern Approach*. Upper Saddle River, N.J. :Prentice Hall, 2010.
- Scutari M, Denis J (2021). *Bayesian Networks with Examples in R, 2nd edition*. Chapman and Hall, Boca Raton.